OPIOIDS AND RAT ERYTHROCYTE DEFORMABILITY

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ABSTRACT

In previous studies from this laboratory, it was noted that opioids in vitro reduced human red blood cell deformability. The effect was found to be dose-dependent, naloxone reversible and preferentially selective kappa ligands exhibited the highest potency. To extend these findings studies were carried out using rat erythrocytes. The time required for erythrocytes to pass through a 5.0 um pore membrane was determined and used as an index of deformability. Opioids added in vitro produced inhibition of deformability in a dose-dependent, naloxone reversible manner. Injecting naive animals with morphine or nalbuphine also produced dose related reductions in red cell deformability. The degree of inhibition produced by nalbuphine correlated well with its plasma concentrations as measured by high performance liquid chromatography (HPLC). Chronic morphine treatment by pellet implantation resulted in-the development of tolerance as evidenced by a loss in the ability of morphine in vitro to inhibit red cell deformability. Addition of naloxone resulted in a decrease in filtration time. Thus, the data confirm and extend previous findings on human red blood cells. In as much as previous data from this laboratory demonstrated that opioids inhibit calcium flux from erythrocytes by inhibiting calcium-ATPase and calcium efflux is necessary for normal deformability, it is concluded that opioids act to reduce red cell deformability by inhibition of the calcium pump.

INTRODUCTION

Opioids cause a reduction in the ability of human red blood cells to deform in a dose dependent, naloxone reversible, stereoselective manner. Relatively selective kappa ligands are more potent in eliciting this effect and extended opioid incubation with erythrocytes produces tolerance and a rebound phenomenon upon exposure to naloxone (Rhoads et al.1985). To extend these findings studies were carried out using rat erythrocytes.

MATERIALS AND METHODS

Heparinized blood was obtained from Sprague Dawley rats (200-300g) under CO_2 anesthesia by cardiac puncture. The time necessary for 1 ml of blood to pass through a 5 um pore Nucleopore membrane under 5 ml water pressure was measured and used as an index of the ability of the erythrocytes to deform

(Reid et al 1976).

The effects of chronic opioid treatment on deformability were studied by making rats morphine tolerant and dependent. One 75 mg morphine-base pellet was implanted on day 0 and 2 pellets on days 1 and 2. Pellets were removed on day 3 and with some rats naloxone was either injected intraperitoneally (5 mg/kg) or added in vitro (0.2 uM) immediately afterwards.

In order to study the correlation between drug concentration and deformability, rats were injected with nalbuphine HCl intraperitoneally and the plasma concentration of nalbuphine was determined according to the HPLC method described by Lo et al. (1984). Nalbuphine was extracted from plasma and separated using a C8, 5 um (4.6 x 250 mm) column, with a mobile phase of 27% CH_3CN, 0.035% H_3PO_4 at a flow rate of 1.4 ml/min using an amperometric detector.

RESULTS

All opioids used in the study inhibited rat erythrocyte deformability dose-dependently. Dynorphin A(1-13) and 1-pentazocine showed the highest potency, the IC_{50} being 0.3 nM and 0.5 nM respectively (figure 1). The lowest potency was observed for D-ala-D-leucine enkephalin (800 nM) and d-pentazocine (925 nM).

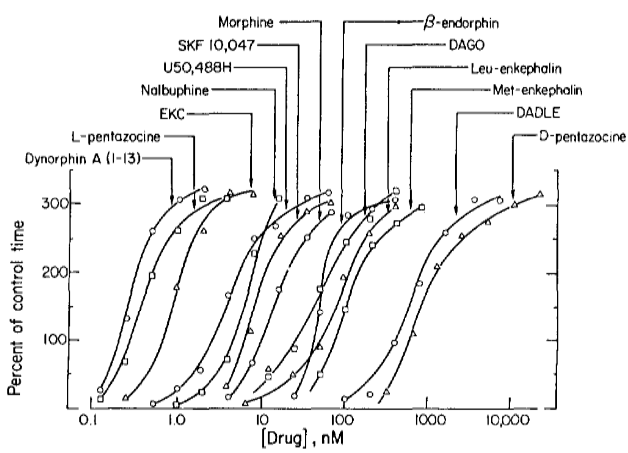


FIGURE 1. Dose response curves of opioids increasing erythrocyte filtration time. Erythrocytes were incubated with opioids in vitro at 37°C for 15 minutes prior to the measurement of filtration time.

In erythrocytes obtained from morphine tolerant/dependent rats, morphine added in vitro exhibited reduced ability to inhibit deformability when compared to those from naive animals (figure 2). An increasing shift of the dose response curve to the right occured with increasing pellet implantation time and when the pellets were removed after the third day of implantation the curves shifted back toward the naive state (figure 2). Addition of naloxone resulted in a rebound effect with an increased deformability of the red cells, as evidenced by a reduction in the filtration time to one half (data not shown).

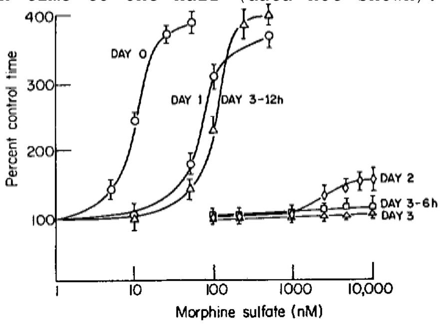
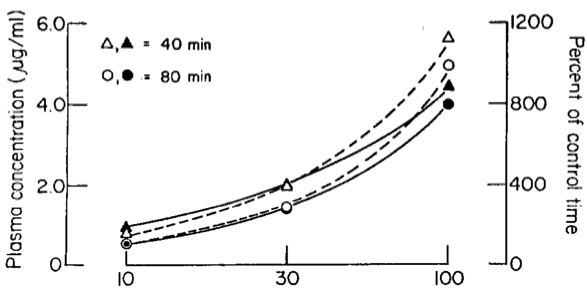


FIGURE 2. Chronology of development and loss of tolerance to morphine inhibition of erythrocyte deformability. Rats were implanted with one pellet of morphine base on day 0, two pellets on days 1 and 2. Blood was withdrawn and the filtration time of the erythrocytes in the presence of varying concentrations of morphine was measured on day 0, 1, 2, 3 and on day 3, six and 12 hours after pellet removal.

HPLC measurement of the plasma concentrations of nalbuphine yielded values that correlated directly with the degree of inhibition of rat erythrocyte deformability (figure 3).



Dose (mg/kg)

FIGURE 3. Nalbuphine plasma concentrations (left y-axis, open symbols and dotted lines) and changes in red cell deformability (right y-axis, closed symbols and solid lines) 40 and 80 minutes after 10, 30, and 100 mg/kg intraperitoneally injected nalbuphine HCl.

DISCUSSION

Our results demonstrate that opioids inhibit rat erythrocyte deformability in a dose dependent, stereoselective manner with relatively selective kappa ligands exhibiting the highest potency. Chronic treatment of rats with morphine resulted in tolerance and a rebound effect occured upon addition of naloxone. These results extend and confirm similar results with human erythrocytes (Rhoads et al. 1985). The HPLC measurements revealed that the deformability changes correlated well with the amount of opioid available in plasma; higher nalbuphine plasma concentrations produced a higher inhibition of deformability.

The change of deformability in erythrocytes is often postulated as a symptom for an existing pathological condition and many agents can increase or decrease the ability of erythrocytes to deform. Intracellular calcium is believed to play a piv tol role in the deformability of red cells and because Yamasaki and Way (1983) found that the calcium-ATPase of rat red cells can be inhibited by opioids, we suggest that such an action inhibits calcium efflux and decreases erythrocyte deformability.

Endogenous peptides with opioid activity have been reported to be released during stress and naloxone to be efficacious in reversing some of debilitating effects of stress (Bernton et al.1985). Since reduced red cell deformability is known to be deleterious to systems by decreasing blood flow, we are presently examining what role a reduced red cell deformability might play under these circumstances.

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